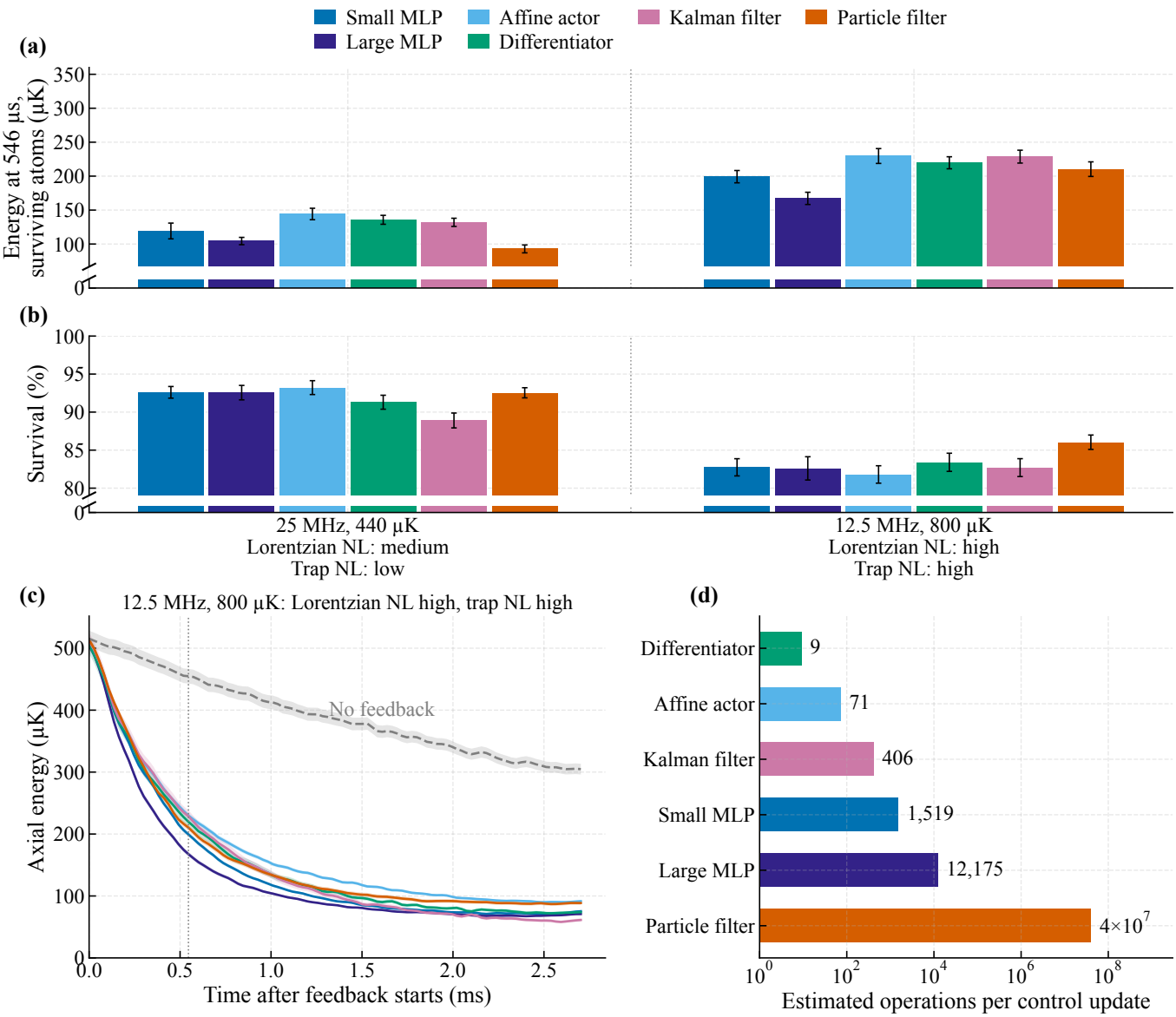

Small MLP
Large MLP
Affine actor
Differentiator
Kalman filter
Particle filter
(a)
Energy at 546 µs, surviving atoms (µK)
(b)
Survival (%)
25 MHz, 440 µK
Lorentzian NL: medium
Trap NL: low
12.5 MHz, 800 µK
Lorentzian NL: high
Trap NL: high
(c)
12.5 MHz, 800 µK: Lorentzian NL high, trap NL high
No feedback
Axial energy (µK)
Time after feedback starts (ms)
(d)
Differentiator 9
Affine actor 71
Kalman filter 406
Small MLP 1,519
Large MLP 12,175
Particle filter 4×10^7
Estimated operations per control update